Pulse-driven ordinary and bifilar coils

When do one-sign pulses build a coil into a large standing wave? — CLAUDE.md §3e step 3

kronspice simulation report · 2026-11-07 · 100 V, 50 Ω one-sign pulses; 20 sections per coil

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Summary

Four coils, each wound from about 126 m of the same 0.5 mm copper wire, are driven by one-sign 100 V pulses through a 5-turn primary: an ordinary solenoid, a Tesla bifilar solenoid (US 512,340), a bifilar cone and a bifilar flat spiral. Each coil is a network of 20 magnetically coupled sections, not a lumped L and C. The example asks at which pulse rates the energy builds up into a large standing wave, and then takes power off the ordinary coil with a low-voltage, high-current pickup.

- **Build-up only in step with resonance.** A one-sign train at rate r contains r, 2r, 3r, …, so a coil builds up only when one of these lands on its first resonance f_1: at r = f_1, $f_1/2$, $f_1/3$. At 0.9, 1.1, 1.5 or 2 × f_1 none does.
- **Ordinary solenoid:** f_1 = 1.167 MHz (wire = 0.489 wavelengths, a free half-wave line), Q ≈ 210; pulses at f_1 give 7,426.4 V between the coil's ends from 100 V pulses (×74).
- **Bifilar windings resonate 28 to 33 times lower** (34.93–41.61 kHz) because of the 12 nF between their strands, with Q ≈ 83–96, and reach 154.6–281.6 V at f_1.
- **V/2 between neighbouring turns.** At r = f_1 neighbouring turns differ by 141.3 V in the bifilar solenoid, 77.9 V in the cone and 127.0 V in the spiral (half the coil voltage), against 37.9 V in the ordinary solenoid.
- **Every joule comes from the pulse source.** At f_1 the ledger closes to 1.6×10^{-5} J (ordinary) and 10^{-6}–10^{-7} J (bifilar).
- **The pickup converts voltage to current; it adds no energy.** A tuned 4-turn pickup at the coil's middle delivers 3.828 A at 0.766 V (1.4010 W), 66.5 % of the power that reaches the primary past the source's 50 Ω; the best variant reaches 91.3 %.

What is being tested

CLAUDE.md §3e step 3 states that, for pulse drive, a coil's reflected wave adds to the drive when the pulse repeats at the round-trip time of the winding (or a whole multiple of it). Tesla's US 512,340 bifilar winding joins the end of one strand to the start of the other, so neighbouring turns sit half a wire length apart in potential (the patent's V/2). This example checks both statements on full distributed models and books the energy of every run. It is standard circuit physics; no hypothesis is involved. The bifilar solenoid's first mode is also the load used by *water_lamp*, and the same coil formulas serve *coiled_tether* and *bial_bifilar*.

The model

Wire and geometry

0.5 mm copper at 0.7 mm between wire centres (0.2 mm of enamel, ε = 3 declared). Each winding holds about the same length of wire as the 400-turn solenoid:

Coil	Turns	Radius	Height	Wire	Surroundings
Ordinary cylindrical solenoid	400	5.0 cm	27.9 cm	125.7 m	grounded can of twice its radius
Tesla bifilar solenoid (US 512,340)	400	5.0 cm	27.9 cm	125.7 m	same can
Tesla bifilar cone, 45° opening upward	248	2.0 to 14.2 cm	12.2 cm	126.4 m	grounded plane 5 cm below the base
Tesla bifilar flat (Archimedean) spiral	214	2.0 to 16.9 cm	0	127.1 m	grounded plane 5 cm below

In a bifilar coil the wire runs out along the even positions (strand 1) and back along the odd ones (strand 2), the end of the first joined to the start of the second, so neighbouring turns belong to the two strands.

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Distributed network

- **Inductance:** 20 sections per coil. The section inductance matrix is summed turn by turn from Maxwell's formula for coaxial circles (*Treatise* art. 701, complete elliptic integrals), with a single turn's high-frequency self-inductance $\mu_0 R(\ln(8R/r_w) - 2)$ (Grover 1946). Sections are coupled in kronspice with K elements.
- **Capacitance:** between adjacent turns, the two-wire formula $\pi \cdot \varepsilon_0 \varepsilon_r \cdot \pi D/\operatorname{acosh}(p/d)$ with each pair's mean diameter D (12.1 nF summed over all neighbouring turns); to the surroundings, the winding as a sheet facing the nearest grounded surface, split half to each end node of a section.
- **Losses:** Terman's skin-effect resistance at f_1; a 1 GΩ bleeder gives the floating coil a DC reference.
- **Drive:** 100 V, 50 Ω source into a 5-turn primary of 1 mm wire at 2 mm pitch just below the base, coupled to every section by the same coaxial-circle mutuals. Pulse edges are 2 % and width 10 % of $1/f_1$; rates r/f_1 = 1/3, 1/2, 0.9, 0.97, 1, 1.03, 1.1, 1.5 and 2; each run lasts $\min(600, \max(60, 5Q/\pi + 20))$ cycles of f_1 (about five ring-up time constants) at 50 steps per period.
- **f_1 and Q** come from an AC sweep of the terminal voltage per volt of drive, 1 kHz to 100 MHz.
- **Declared approximations:** sections are equipotential inside; the bifilar junction lead and radiation are left out; proximity effect is not in the wire resistance; absolute tolerances are loosened to the circuit's scale (1 µA, 1 mV, 1 pC).

Results

The coils

Coil	f_1	Wire / λ	AC gain at top	Q	Inductance	Pulse width	Cycles
Ordinary solenoid	1.167 MHz	0.489	×338.6	210	4.853 mH (Nagaoka 4.873 mH)	85.7 ns	354
Bifilar solenoid	41.61 kHz	0.017	×12.1	83	4.853 mH	2.40 µs	151
Bifilar cone	35.81 kHz	0.015	×6.6	96	6.542 mH	2.79 µs	172
Bifilar flat spiral	34.93 kHz	0.015	×10.8	94	6.842 mH	2.86 µs	169

The two solenoids have the same inductance to 10^{-12} (the strands of a series bifilar coil carry the same current the same way), within 0.4 % of Nagaoka's current-sheet formula; the flat spiral is within 1.7 % of Wheeler's flat-spiral formula. The ordinary solenoid's wire is 0.489 wavelengths at f_1: it rings as a free half-wave line, the first resonance Knight reports for a free single-layer coil, with voltage maxima at its ends and the current maximum at its middle.

Pulse-rate sweep

Voltage between the coil's ends at the end of each run (100 V pulses from 50 Ω):

Coil	$f_1/3$	$f_1/2$	0.9 f_1	0.97 f_1	f_1	1.03 f_1	1.1 f_1	1.5 f_1	2 f_1
Ordinary solenoid	2631.5 V	3868.1 V	454.8 V	892.8 V	**7426.4 V**	657.4 V	213.9 V	479.5 V	356.4 V
Bifilar solenoid	97.4 V	143.4 V	15.9 V	54.9 V	**281.6 V**	58.5 V	19.5 V	6.6 V	5.1 V
Bifilar cone	53.4 V	78.6 V	7.9 V	26.5 V	**154.6 V**	28.4 V	9.9 V	3.9 V	3.1 V
Bifilar flat spiral	86.8 V	127.9 V	12.8 V	43.6 V	**252.1 V**	46.3 V	15.5 V	5.7 V	4.7 V

Voltage between neighbouring turns at r = f_1: 37.9 V (ordinary), 141.3 V (bifilar solenoid), 77.9 V (cone), 127.0 V (spiral). Every bifilar winding has half its coil voltage between neighbouring turns, as the patent says.

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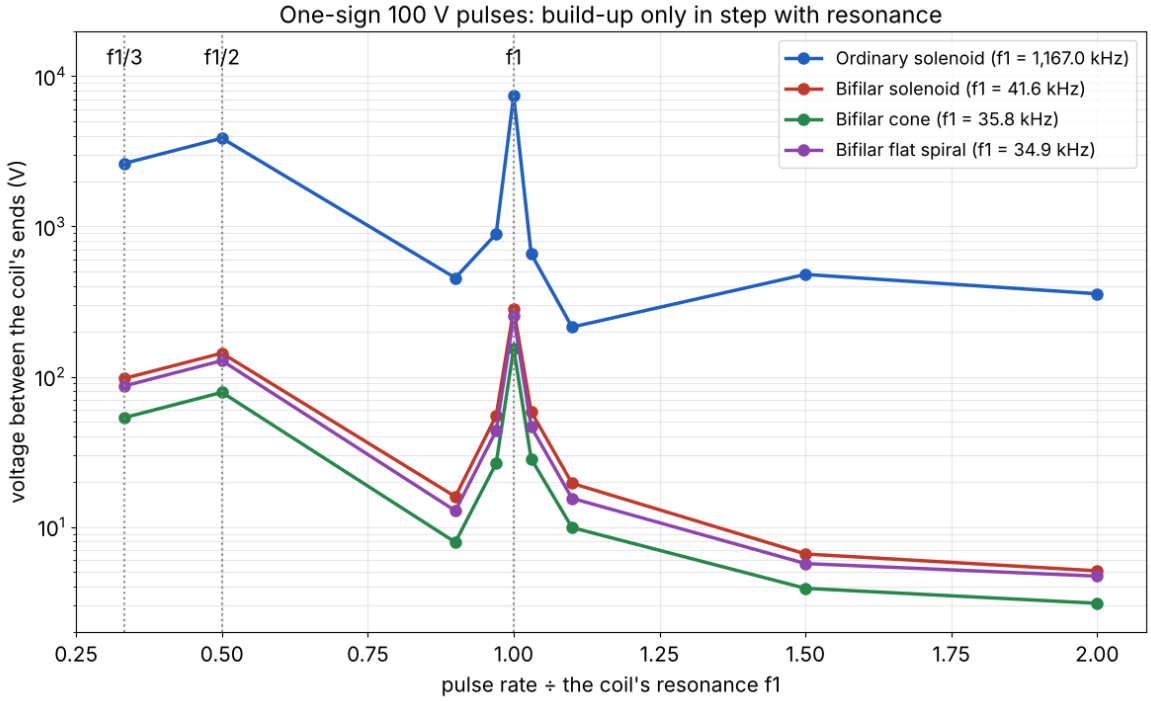


Figure 1. Voltage between the coil's ends against pulse rate ÷ f_1 (log scale), for the four coils. Peaks appear only at $f_1/3$, $f_1/2$ and f_1, where one harmonic of the one-sign train lands on the resonance.

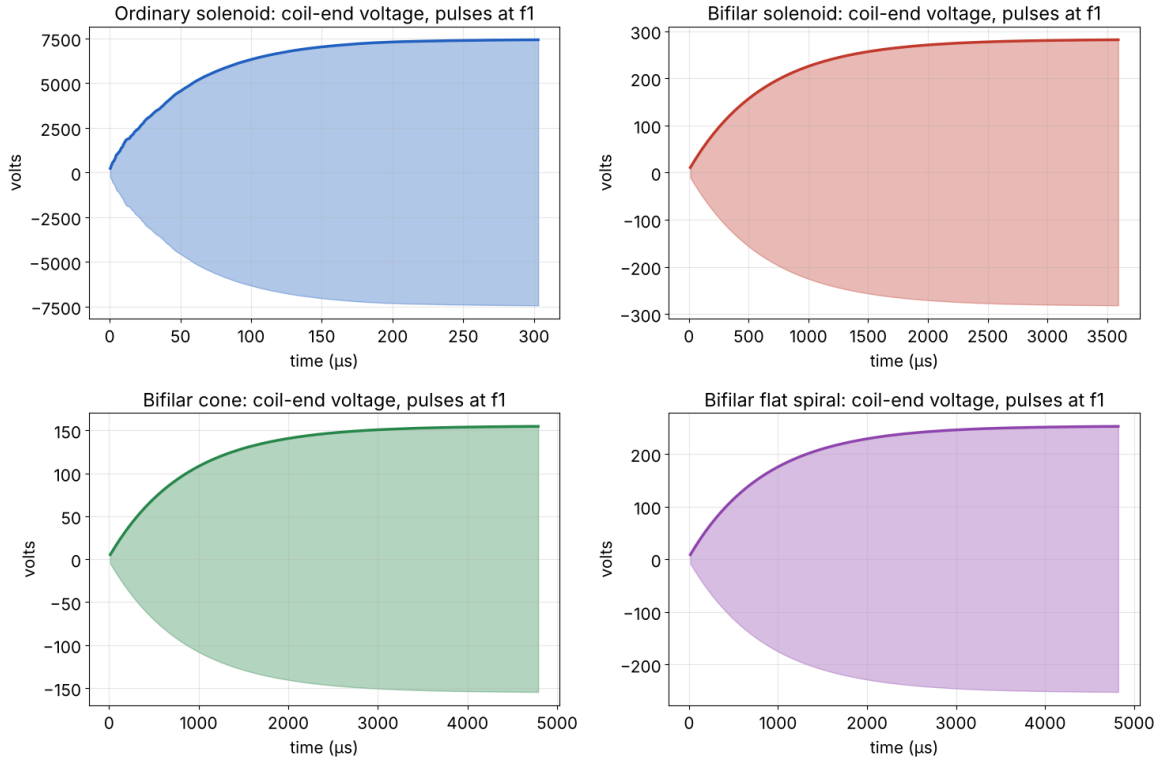


Figure 2. Coil-end voltage envelope building up pulse by pulse with pulses at f_1. The ordinary solenoid settles near 7.4 kV within about 300 µs; the bifilar coils settle at 150–280 V over a few milliseconds.

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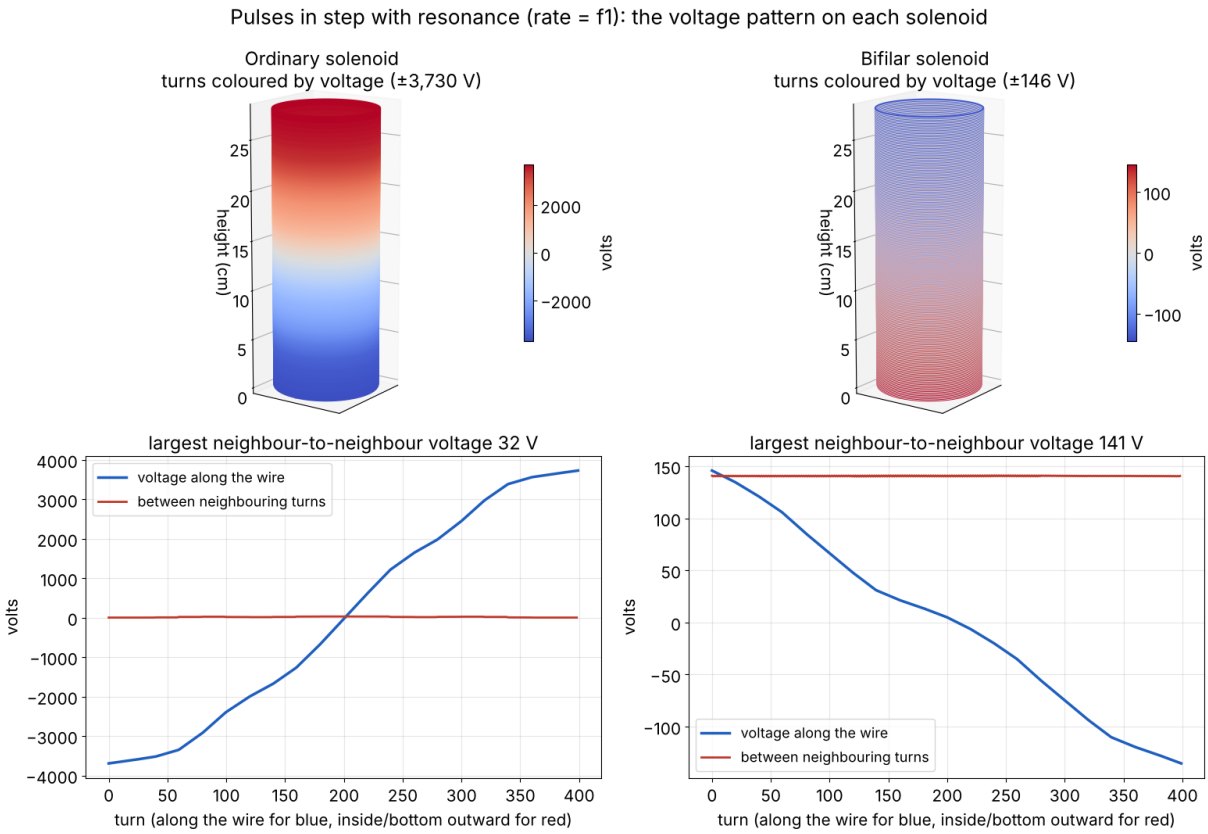


Figure 3. The two solenoids at f_1, at the moment of largest voltage. Top: every turn coloured by its voltage. Bottom: voltage along the wire (blue) and between neighbouring turns (red). The ordinary coil shows a half-wave pattern with a small turn-to-turn voltage (32 V at this instant); in the bifilar coil every neighbouring pair differs by about half the coil voltage (141 V).

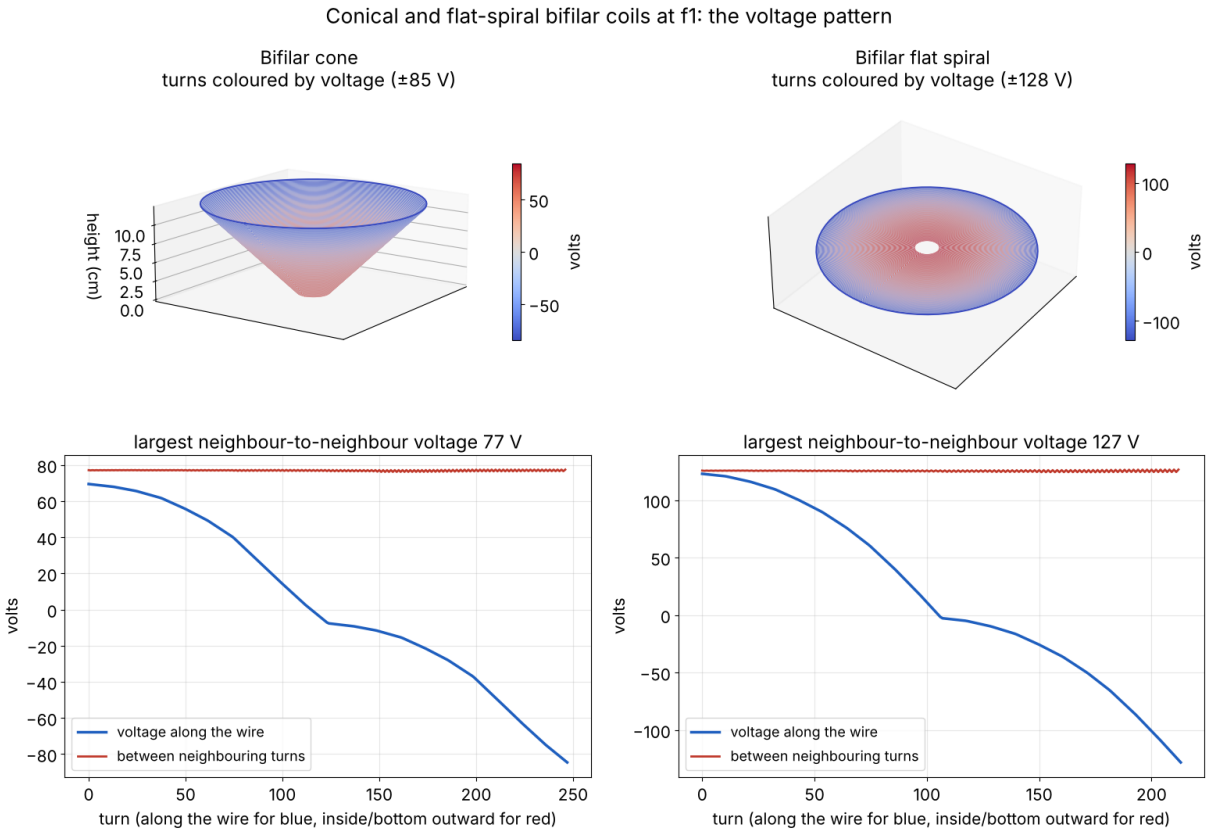


Figure 4. The bifilar cone and flat spiral at f_1, drawn as in Figure 3. Neighbouring turns differ by 77 V (cone) and 127 V (spiral), about half the end-to-end voltage.

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Low-voltage, high-current pickup

A 4-turn pickup of 2 mm copper (radius 6.5 cm, inside the can; L = 4.082 µH, 0.072 Ω at f_1) is wound round the ordinary coil's middle, where the current is largest, and tuned with a series capacitor $C = 1/((2\pi f_1)^2 L) = 4.56$ nF. The load (and, where stated, the capacitor) is chosen from a grid for the largest load power per V² of drive; the pulse rate is the coupled system's best frequency, searched from 0.7 to 1.4 f_1. "Past 50 Ω" is the power that reaches the primary after the source's own resistance; the share is load power divided by it.

Pickup	C (nF)	Load (Ω)	Rate / f_1	I load (A)	V load (V)	P load (W)	Past 50 Ω (W)	Share	Coil (V)
Middle, tuned $1/(\omega_1^2 L)$	4.56	0.2	1.2975	3.828	0.766	1.4010	2.1055	66.5 %	3651
Middle, capacitor optimized	1.14	10	0.9717	0.667	6.668	1.1996	1.3138	91.3 %	3442
Middle, untuned	none	1	1.1073	1.702	1.702	1.3051	1.6369	79.7 %	3502
Top (high-voltage end), tuned	4.56	0.5	1.0487	2.038	1.019	1.0064	1.1590	86.8 %	2230

- The pickup's load never gets more than the power that reaches the primary past the source's resistance; the pickup converts voltage into current and does not add energy.
- A tightly coupled tuned pickup splits the coil's resonance in two (about 0.83 and 1.30 f_1), so the best pulse rate moves with it (1.2975 f_1 for the tuned middle pickup).

Energy ledger

Energy from the pulse source over each run at $r = f_1$, split into heat and energy stored at the end:

Coil, at f_1	In	Heat	Stored	Time-stepping error
Ordinary solenoid	0.0023518 J	0.0022037 J	0.00013172 J	1.6×10^{-5} J
Bifilar solenoid	0.079822 J	0.079705 J	0.0001151 J	1.2×10^{-6} J
Bifilar cone	0.1083 J	0.10827 J	3.4799×10^{-5} J	1.4×10^{-7} J
Bifilar flat spiral	0.10853 J	0.10844 J	9.2976×10^{-5} J	9.9×10^{-8} J

The bifilar coils take far more energy from the source (their wide pulses at low f_1 spend most of it as heat in the primary circuit) and store less: their low-frequency, low-Q mode builds up less than the ordinary coil's half-wave resonance. Pickup ledgers:

Pickup	In	Heat (of it, load)	Stored	Time-stepping error
Middle, tuned	0.003514 J	0.003469 J (0.0003756 J)	4.23×10^{-5} J	2.7×10^{-6} J
Middle, capacitor optimized	0.002382 J	0.002349 J (0.0003185 J)	2.72×10^{-5} J	3.4×10^{-6} J
Middle, untuned	0.00323 J	0.003187 J (0.0003571 J)	2.85×10^{-5} J	1.5×10^{-5} J
Top, tuned	0.002516 J	0.002493 J (0.0002707 J)	2.21×10^{-5} J	6.5×10^{-7} J

Tests

Test	Checks
TestCoilNetwork	every winding uses each position once; L within 1 % of Nagaoka and equal for both solenoids; symmetric capacitance matrix; the bifilar coil holds (N − 1) strand-to-strand gaps
TestFlatSpiralInductance	the spiral against Wheeler's $a^2N^2/(8a + 11c)$, held to 10 %
TestPulseBuildUp	pulses at f_1 build more than three times higher than at 1.5 f_1; ledger closes to 1 %
TestPickup	the tuning capacitor resonates the pickup at f_1 to 10^{-12}; ledger to 1 %; load power below the power past the source
TestExport	the four coils swept around resonance, and the tuned pickup, as netlists and results sheets

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Limits of the model

- Twenty sections per coil, each equipotential inside; finer structure along the wire is not resolved.
- The bifilar junction lead and radiation are not modelled; proximity effect is not in the wire resistance.
- Turn-to-turn capacitance uses the two-wire formula; capacitance to the surroundings treats the winding as a sheet.
- Absolute tolerances are loosened to the circuit's scale (1 µA, 1 mV, 1 pC).

Reproducing the results

```
go run ./examples/pulsed_bifilar                    # about 10 minutes, writes viz/*.csv
go run ./examples/pulsed_bifilar -only-pickup       # the pickup study alone
python3 examples/pulsed_bifilar/viz/render.py       # pictures and animations
go test ./examples/pulsed_bifilar
KRONSPICE_EXPORT=1 go test ./examples/pulsed_bifilar -run TestExport
```

Sources: N. Tesla, US 512,340 (1894) and US 593,138 (1897); J. C. Maxwell, *Treatise* Vol. II art. 701; F. W. Grover, *Inductance Calculations* (1946); H. Nagaoka, *J. Coll. Sci. Tokyo* 27, 1 (1909); H. A. Wheeler, *Proc. IRE* 16, 1398 (1928); Ramo, Whinnery & Van Duzer, ch. 4; F. E. Terman, *Radio Engineers' Handbook* (1943); D. W. Knight, self-resonance of solenoid coils (2016), §13.

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